Problem 1

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Problem 1 *Solve the following word problems:*

(a) *The velocity function for an object moving along a line east/west is given by $v(t) = -t^2 + 4t - 3$ feet per minute.*

(i) *Find the total displacement the object traveled from 2 minutes to 6 minutes (assume east is positive).*

(ii) *Find the total distance the object traveled from 2 minutes to 6 minutes.*

(iii) *Suppose that the object's position 2 minutes into the trip is 5 feet of a placement marker. What is its position (relative to the placement marker) at 6 minutes.*

(b) *Sammy the Snail sets up camp in the median of I-70 and, starting at noon and ending at 6pm, hikes back and forth along the highway. He starts his hike at his campsite. His velocity at time t hours (after noon) is given by $v(t) = (t - 2)(t - 5)$ inches per hour. Find the total distance Sammy travelled on his hike.*